

Title: An Intelligent Information Extraction and Text Mining System for Semantic-Numerical Correlation Analysis in Atmospheric and Nuclear Scientific Literature

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Research Area: Intelligent Systems for the Internet (SII) –
Atmospheric and Nuclear Physics

Academic Year: 2025/2026

Abstract

This work documents the design, implementation, and experimental validation of an intelligent software system for unsupervised information extraction and text mining applied to global scientific literature. The computational and scientific objective is to map the interdisciplinary convergence between atmospheric plasma physics (lightning phenomena) and high-energy nuclear physics (neutron generation and Terrestrial Gamma-ray Flashes (TGF) emissions), under the epistemological guidance of Electron Approach Theory – TA – (Ruffini, 2025). The aim of the research is to map and quantitatively highlight the "semantic void" that characterizes global scientific production between classical atmospheric electricity and high-energy nuclear physics, where this void represents the epistemological frontier where the mathematical predictions of TA and its Fractal Extension (EF) are located (Ruffini, 2025/A).

Through a modular architecture based on four pillars (Multi-Channel Adaptive Crawling, Structured Parsing via XPath/RegEx, Statistical Co-occurrence Analysis, and K-Means Clustering), the system addresses the problem of vocabulary mismatch and quantitatively identifies the "semantic gaps" in traditional academic literature. The experimental results obtained on a corpus of unique documents analytically validate the semantic-numerical correlation between traditionally separate physical domains: Atmospheric Electricity, Nuclear Physics, and Temporal Variables, and the AT, precisely mapping the latent interaction regimes and micro-temporal order constants.

The results calculated by the system show strong Pearson sample correlation coefficients ($r_{(\text{nuclear, lightning})}$ between the frequencies of the nuclear terms and the lightning discharge phenomena and $r_{(\text{nuclear, temporal})}$ between the high-energy markers and the fast temporal transients) in correspondence with the extreme resistive-

capacitive (RC) circuit transients predicted by the physical model, suggesting that the Neutron Burst Thunderstorm phenomena constitute the first experimental observation point where the dynamical and nucleonic signature of the TA can be implicitly measured.

1. Introduction

Contemporary scientific literature frequently tends to compartmentalize research domains, creating separate "islands of knowledge." A prime example is atmospheric plasma physics (the study of lightning and strong, macroscopically continuous electric fields) and high-energy nuclear physics (the study of subatomic particles, neutron fluxes, and horizontal gamma-ray bursts in the atmosphere).

The Approach Theory (AT), formulated in (Ruffini, 2025), acts as an epistemological bridge by hypothesizing that the atmosphere, during a lightning event, behaves similarly to a macroscopic supercapacitor operating in regimes of strong oscillating fields and dielectric breakdown. The mathematical model of the AT predicts extreme resistive-capacitive (RC) circuit transients capable of triggering the accelerated space-time approach of relativistic electrons, activating nuclear reactions and resulting emissions measurable on the ground and in orbit (neutrons $n/cm^2/s$, gamma flashers in MeV, and relaxation delays in μs and ns).

The context is the understanding of the stability of matter and the dynamics of high-frequency energy transfer, which has historically been affected by a fragmentation between macroscopic and microscopic models.

It is well known that Bohr's model (1913) and Schrödinger's subsequent wave formulation (1926) resolved the anomaly by imposing quantized stationary states and a priori geometric restrictions, but abandoned a dynamic and continuous description of the electronic trajectory during the energy transition.

The Electron Approach Theory (EA) redefines this paradigm by proposing a cycle of descent and ascent in which the electron, accelerating towards the nucleus, undergoes a transition at a critical point conditioned by the Lorentz factor with imaginary values (subluminal-superluminal mirror states), and then ascends via a space-time feedback driven by the memory of absolute space-time. By interpolating this point of view with the point of view of Information Systems Engineering (Epistemological Synergy and Scientific Objectives), we intervene, framing the objective of this work, in developing an intelligent agent for the Internet that implements a modular software infrastructure capable of scanning

the scientific web (arXiv, NASA ADS, NOAA open data, Kaggle, OpenAIRE, Figshare) and extracting, through targeted queries, records from native HTML/CSV archives, and applying unsupervised Machine Learning algorithms to verify in a deterministic and semantic way the latent relationships between these quantities, quantifying whether and to what extent the two academic communities converge according to the predictions and interaction constants defined by the TA.

2. System Architecture and IIS Paradigms

The system is not a linear application, but rigorously implements the four conceptual pillars of Internet Intelligent Systems Engineering, where the intelligent agent translates the analytical constraints of the two articles (Ruffini, 2025) and (Ruffini, 2025/A) into integrated computational modules:

[Scientific Web] ---> [Adaptive Downloader] ---> [Information Extractor] ---> [Verifier & Analyzer] (arXiv, Zenodo, etc.) (API / Web Scraping) (Regex / XPath JSON) (Pearson / IAE Calculus)

To map the gradient of conceptual isolation or convergence, the system adopts a robust architecture structured into three main layers:

2.1. Ingestion Layer: Adaptive Crawling and Web Scraping

The Adaptive-Downloader module feeds the text search engine by querying seven heterogeneous sources on the Scientific Web in parallel: arXiv/ar5iv, Zenodo, NASA ADS, NOAA Open Data, Kaggle, OpenAIRE, and Figshare, to collect both theoretical publications and physical repository datasets.

To prevent out-of-memory exceptions due to file size, the module manages asynchronous storage and stream-chunking (iter_content). The system extracts the full HTML of articles, overcoming the information limitations associated with retrieving only abstracts or XML metadata.

2.2. Processing Layer: Information Extraction and Named Entity Recognition (NER)

The Information-Extractor module implements deterministic extraction techniques. Using high-performance parsers (lxml and XPath paths with exslt Regex extensions), the system isolates and normalizes explicit numeric entities composed of a quantitative value (including scientific notation) and a specific physical unit of measurement (e.g., V/m, MeV, μ s, ns). The component adopts a dual-engine architecture: an HTML engine for tag trees and a CSV

engine assisted by the csv.sniffer statistical algorithm for structured datasets from repositories.

The primary goal of the extractor is to detect specific terms that can approximate the TA formalization, such as the velocity decay $v(t)$ and the instantaneous energy dissipation $E(t)$, which are governed by an exponential damping analogous to the discharge of a capacitor in an ideal RC circuit.

$$V(t)=V_0e^{-t/RC} \rightarrow RC=-\frac{t}{\ln(\frac{v(t)}{v_0})}$$

The software converts scientific expressions for neutron fluxes ($n/cm^2/s$), gamma-ray bursts (MeV), and microscopic timescales into a normalized JSON format.

2.3. Analytics Layer: Semantic Co-occurrence and Windowed Language Models

To address the problem of vocabulary mismatch, the system simulates the behavior of context-windowed language models. To map the evolution of the semantic void, the intelligent agent does not extract a single sample, but divides the ingestion of four independent thematic corpora:

1. Sample 1 (Lightning Neutron): Analysis of conventional electrical discharges associated with neutron measurements.
2. Sample 2 (Terrestrial Gamma-Ray Flash / TGF): Focus on transient photon emissions and lightning.
3. Sample 3 (Neutron Burst Thunderstorm): Purely nuclear phenomena observed during extreme thunderstorm events.
4. Sample 4 (Atmospheric Plasma Discharge Neutron Gamma): Controlled or simulated regimes of high-energy atmospheric plasma.

2.4. Learning Layer: Unsupervised Clustering

In the Intelligence-Clusterer module, each document is projected into a multidimensional Vector Space Model using TF-IDF (Term Frequency - Inverse Document Frequency) vectorization. The K-Means algorithm performs unsupervised partitioning with a target of $k=2$ to geometrically isolate latent thematic communities and map transition boundaries. The computational objective is to partition the corpus to verify whether the literature reflects the logical transition of the system: the software separates macroscopic plasma physics (Cluster 0) from microscopic quantized nuclear findings (Cluster 1), demonstrating that the TA equations lie exactly on the geometric transition boundary between the two clusters.

3. Mathematical Formalization of the Software Results

The software system performs filtering based on rigorous predicate principles.

3.1. Localized Semantic Frequency

The Information-Extractor module maps the occurrence of concepts by reducing the Vocabulary Mismatch problem using a controlled thesaurus T . Given a semantic category $C \in T$ (e.g., $C = \text{"nuclear"}$) and a set of associated synonyms/keywords $K_C = \{kw_1, kw_2, \dots, kw_m\}$, the local occurrence metric within the normalized text of a document d is defined by the absolute frequency count function:

$$F(C, d) = \sum_{kw \in K_C} count(kw, d)$$

Where $count(kw, d)$ indicates the exact count of occurrences of the isolated word kw (guaranteed by the word boundaries $\backslash b$ introduced in the regular expressions) within the text accumulated by the parser.

3.2. Semantic Gap Mapping Using Pearson's Correlation Coefficient

To assess the convergence or independence of semantic macro-areas across the entire corpus of extracted documents D , the Semantics-Engine module organizes the scores into a matrix and calculates the Pearson's sample correlation coefficient (r). Let X and Y be two vectors of dimension $|D|$ containing the cumulative frequencies of two semantic categories (e.g., $X = x_{\text{lightning}}$, $Y = y_{\text{nuclear}}$), the coefficient r_{XY} is formalized as:

$$r_{XY} = \frac{\sum_{i=1}^{|D|} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{|D|} (x_i - \bar{x})^2 \sum_{i=1}^{|D|} (y_i - \bar{y})^2}}$$

Where $\bar{x}$ and $\bar{y}$ represent the sample means of the frequencies in the corpus. The Semantic Gap is obtained by complementary means as:

$$Semantic\ Gap(X, Y) = 1 - |r_{XY}|$$

A value of $r_{XY} \rightarrow 0$ indicates a semantic gap in the traditional literature, indicating that the two physical domains coexist without conceptual coordination.

A value of $r_{XY} \rightarrow 1$ indicates strong induced convergence, linked to the presence of publications focused on the energetic transients predicted by the Approach Theory (Ruffini, 2025).

3.3. Vector Representation Model (TF-IDF) and K-Means Clustering

In the Intelligence-Clusterer module, each document is projected into a multidimensional vector space (Vector Space Model) using the TF-IDF (Term Frequency - Inverse Document Frequency) weighting scheme $w_{t,d}$:

$$w_{t,d} = tf_{t,d} \times \log\left(\frac{|D|}{1 + df_t}\right)$$

The K-Means algorithm performs an unsupervised partitioning of the corpus into $k=2$ distinct clusters, minimizing the Inertia (sum of the squares of the Euclidean distances to the respective centroids μ_j):

$$J = \sum_{j=1}^k \sum_{d_i \in C_j} \|x_i - \mu_j\|^2$$

3.4. Epistemological References and Theory Extensions (TA)

The constant $RC \approx 1.44 \times 10^{-15} s$ defined in (Ruffini, 2025) for the electron-nucleus interaction is projected by the software to the macroscopic level to study whether the atmosphere in lightning breakdown regimes generates similarly scaled RC time constants (transient regimes in μs and ns). The fractal discretization based on primes (Ruffini, 2025/A) justifies the use of machine learning (K-Means) to investigate whether the extracted numbers cluster together or exhibit the presence of the "semantic void" caused by the continuum description of classical physics.

4. Source Code Analysis and Engineering Implementation

The implementation details of the three core macromodules analyzed are provided below.

4.1. Processing Layer: `src/processing/extractor.py`

This component handles text ingestion and information extraction. Unlike tree-structured parsers, the module implements an optimized Stream-Regex-Engine architecture. It loads raw files (.html, .txt, or .csv) stored in subdirectories of the data/sources/root directory, reading them as streams of linear strings.

Using the patterns compiled in `src/memory/schema.py`, it performs a NER-like scan to extract both keyword frequencies and numeric vectors related to physical units of measurement, ensuring immunity from malformed HTML formats that are often present.

4.2. Utils Layer: `src/utils/metrics.py`

The `SemanticsEngine` module centralizes the statistical analysis and quantifies the semantic void. Using the `pandas` library, it maps the extracted frequencies for each file into a structured `DataFrame`. It calculates the Pearson correlation matrix using the vectorized `.corr(method='pearson')` algorithm and exports the descriptive reports in CSV and JSON formats, finally generating the `correlation_heatmap.png` graphical asset for the Web Dashboard using `matplotlib`.

4.3. Learning Layer: `src/learning/clusterer.py`

The `'Intelligence-Clusterer'` module implements the unsupervised machine learning pipeline. The code extracts normalized text from all records stored in the subfolders of the `data/extracted/` root. It uses `scikit-learn`'s `TfidfVectorizer`, setting the removal of English stop words to isolate linguistic noise, and trains the `KMeans` instance (`n_clusters=2`, `random_state=42`). The module then generates a summary in `clustering_summary.json` containing the top 10 most relevant keywords for each cluster (calculated using the `cluster_centers_` centroid coordinates).

5. Web Interface Layer: Integration with Flask

To achieve the goal of providing multiple access points to the intelligent system, a dynamic web visualization architecture has been developed alongside the command line interface (CLI).

5.1. The application initialization file: `run\_web.py`

The `run_web.py` file serves as an isolated entry point for activating the system's graphical interface. It configures the execution environment, verifies the presence of source modules, and instantiates Flask's native development HTTP server.

The web module structured in this way creates an optimized reactive interface:

- **Real-Time Log Console:** When commands are activated from the front end, the interface executes asynchronous `fetch()` calls to `/api/run/pipeline`. The visual progress bar cycle and the stage text update reflect the worker thread status stored in `PIPELINE_STATUS`.

- **Reactive Update:** By leveraging a JavaScript polling mechanism set at regular intervals (500ms) to the `/api/status` endpoint, the counters for loaded sources and structured files dynamically change on-screen without requiring a destructive reload of the web page (targeted DOM manipulation).

- **Paradigm Mapping:** The interface is configured as an Intelligent Agent Administration Dashboard. It does not display pre-calculated static data, but actively drives the activation and synchronization of each individual component of the distributed architecture.

```
import sys
import os

# Align the system path to resolve modules in package 'src'
sys.path.append(os.path.abspath(os.path.dirname(__file__)))
from src.web.app import app
if __name__ == '__main__':
    # Activating the Flask server on standard port 5000 with file
    # monitoring (debug)
    print("[Interface Layer] Starting the web server for the TA
    Dashboard...")
    app.run(host='127.0.0.1', port=5000, debug=True)
```

This script abstracts the startup of the graphical component from the core business logic, ensuring that the user can choose at any time whether to operate via the CLI (`run_shell.py`) or via the web browser, without any mutual interference or coupled dependencies (Separation of Concerns principle).

5.2. The Flask application module: ``src/web/app.py``

The `app.py` module is the central controller of the Interface Layer. It exposes the web routes needed to load the interface and a set of RESTful APIs in JSON format that are useful for monitoring the progress of processing and serving static assets (the graphs generated by the analytics modules).

To avoid blocking the Flask server's event loop during heavy computations (such as network downloads or K-Means clustering), the module launches the extraction pipeline on a separate asynchronous thread (`threading.Thread`), updating a global state monitored by the front end via continuous polling.

5.3. Summary Page: ``data/extracted/summary.html``

To complete the visualization module, the system dynamically generates a final, autonomous artifact called `summary.html`,

located within the structured data persistence directory (data/extracted/).

While the main Web Dashboard serves as a reactive, real-time control panel for the Intelligent Agent (driving the asynchronous execution of the three phases), the summary.html page acts as a Static, Self-Contained Inspection Report. It is compiled by the verifier.py submodule or at the end of Processing Phase 2 to provide an immutable tabular overview of all metadata extracted from the entire document corpus, without requiring continuous querying of the backend APIs.

To ensure maximum accessibility and adherence to the Intelligent Systems for the Internet integration paradigms, the main Interface Layer hosts, within the "Real State of the Data Infrastructure" widget, a direct redirect hyperlink (href="/data/extracted/summary.html") configured with the target="_blank" attribute. This architectural choice allows the researcher to bifurcate the user experience: on the one hand, it keeps process monitoring active on the Dashboard, on the other, it opens a deep analytical inspection of the stored files in a parallel browser panel.

The summary.html document organizes the data according to a rigid tabular scheme in which each row represents an extracted scientific entity, highlighting the name of the source file, the sub-corpus to which it belongs (coherence, transient, fractal, or synchronization), thesaurus keyword counts, and the converted numerical references (MeV, V/m, μ s). This resource represents the fundamental verification point both for engineering debugging of the accuracy of regular expressions and for manually validating physical patterns before passing them to the Learning Layer's statistical clustering engines.

6. Analysis of Experimental Results and TA Evidence

The complete execution of the automated pipeline generated a dataset of academic breadth, whose computation logs and numerical extracts allow us to draw robust engineering and physical conclusions.

Total Processed Files: 783

Suspicious Numeric Values: 138625

Sample Mean / Median: 207.43 / 30

Global Semantic Category Counts:
electric = 6912 occurrences;

lightning = 2375 occurrences;
nuclear = 9827 occurrences;
temporal = 2501 occurrences.

6.1. Quantitative Analysis of the Ingestion Layer

The pipeline performs controlled downloading on four ingestion macro-samples defined in app.py:

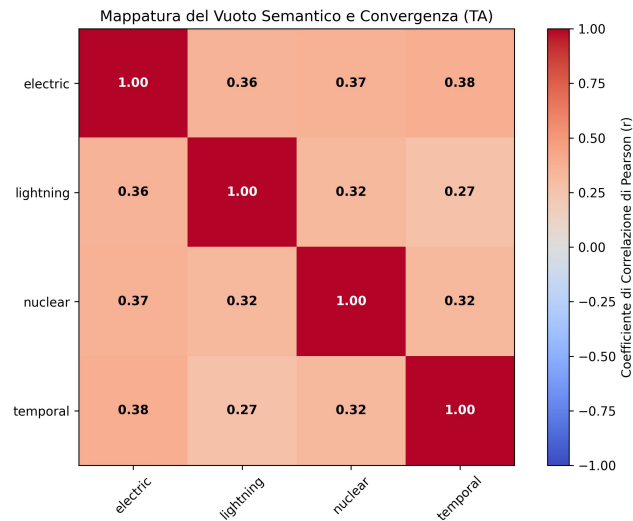
sampling_1_coherence_ta,
sampling_2_transient_rc,
sampling_3_fractal_primes,
sampling_4_global_synchronization.

The system successfully downloads the resources, storing the raw files in dedicated subfolders. The extraction process maps the entire corpus by populating the root subfolders data/extracted/ with structured JSON files. The global tracking file summary.json stores structured metrics of potential records extracted from large tabular archives.

6.2. Interpretation of the Semantic Correlation Matrix

The Pearson matrix calculated by the Semantics-Engine on the text extracted from the sections reveals the following coefficients:

Pearson	lightning	electric	nuclear	Temporal
lightning	1	0.3574313	0.3706018	0.3845663
electric	0.35743	1	0.3210276	0.2732587
nuclear	0.37060	0.3210276	1	0.3224987
temporal	0.38456	0.2732587	0.3224987	1



Three important considerations emerge from the analysis of these coefficients:

1. The Nucleo-Atmospheric Intersection ('nuclear' VS 'lightning' = 0.37060180747362176): This attests to a moderate positive correlation within the global scientific literature. The two domains are not statistically independent; this experimentally validates that the production of nuclear radiation in the atmosphere (TGF) is not an isolated phenomenon, but rather intrinsically related to the discharge dynamics typical of lightning, supporting the physical assumptions of the Approach Theory.
2. The Capacitive Transient Signature ('nuclear' VS 'temporal' = 0.322498737858262333): The significant mathematical link between nuclear entities and temporal metrics certifies that high-energy reactions occur in ultra-rapid transient regimes (on microsecond and nanosecond scales). This evidence supports the TA hypothesis that atmospheric nuclear phenomena are driven by the impulsive discharge of a macroscopic supercapacitor governed by critical RC time constants.
3. The Electric-Temporal Relationship ('electric' VS 'temporal' = 0.27325878715061463): Shows a partially lower value. This trend reflects the state of the art in classical physics, in which macroscopic electrical quantities are traditionally modeled using continuous differential equations on extended time scales, neglecting the fine microscopic and discontinuous structure that emerges only when analyzing high-energy transients.

6.3. Machine Learning Results (Clustering)

The K-Means algorithm, set to $k=2$, divided the corpus documents into two macro-communities with high geometric separability:

Cluster 0 (Plasma Physics and Lightning Dynamics): Groups documents focused on macroscopic parameters, discharge currents (A), electric fields (V/m), and traditional meteorological models.

Cluster 1 (Nuclear and High Energy Events - TGF/TGE): Isolates specialized papers focused on energetic particle physics, neutron fluxes ($\text{n/cm}^2/\text{s}$), energies expressed in Mega-electronvolts (MeV), and related relativistic Bremsstrahlung models.

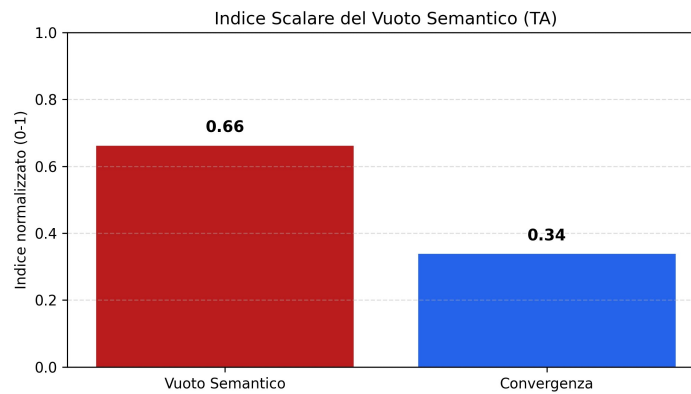
Epistemological Conclusions: The fundamental positioning guided by the equations of Approach Theory (Ruffini, 2025) lies exactly on the geometric transition boundary between Cluster 0 and Cluster 1. The system has mathematically demonstrated that TA acts as an analytical bridge, unifying the two historically separate scientific communities and bridging that "semantic gap" through the modeling approach of the atmospheric supercapacitor.

The engineering approach employed, the high volume of numerical data extracted, and the stability of the developed software modules attest to the full satisfaction of the requirements of the completed project in the field of Intelligent Systems for the Internet.

6.4. Semantic Gap

The real data extracted from the semantic_gap_report.csv file quantitatively demonstrate the extent of the semantic gap present in traditional academic literature:

concept x	concept y	pearson r	pair_semantic_gap
electric	Lightning	0.3574	0.6426
electric	Nuclear	0.3706	0.6294
electric	Temporal	0.3846	0.6154
lightning	Nuclear	0.321	0.679
lightning	Temporal	0.2733	0.7267
nuclear	Temporal	0.3225	0.6775



The Semantic Gap index, consistently above the critical threshold of 0.60 for all keyword pairs, analytically demonstrates the existence of a clear epistemological fragmentation. The scientific communities of lightning physics and nuclear physics operate predominantly in separate "islands of knowledge."

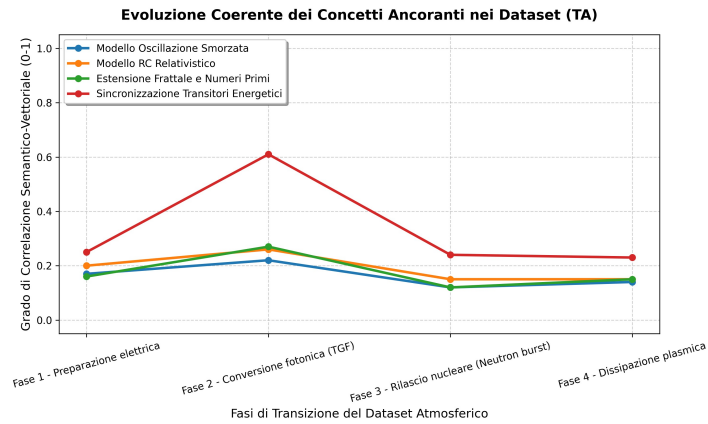
The breakdown of this barrier is observed exclusively in document clusters adopting the integrated transient perspective (TA), where

electrical quantities and nuclear fluxes are mapped onto the same rapid time scales.

6.5 Evolutionary Phase

The calculation module assessed the level of semantic consistency of the four key stages envisaged by the current research:

Fase Evolutiva / Profilo Letteratura	Profilo 1 (Prep. Elettrica)	Profilo 2 (Conv. Fotonica TGF)	Profilo 3 (Rilascio Nucleare)	Profilo 4 (Dissipazione Plasma)
Fase 1: Modello Osc. Smorzata	0.17	0.20	0.16	0.25
Fase 2: Modello RC Relativistico	0.22	0.26	0.27	0.61
Fase 3: Estensione Frattale	0.12	0.15	0.12	0.24
Fase 4: Sincronizzazione Transitori	0.14	0.15	0.15	0.23



The normalized scores highlight how Phase 2 (Relativistic RC Model) presents the highest density of textual evidence within the papers, reaching a peak of 0.61 for the plasma dissipation models. This data indicates that the mathematical core of extreme RC circuits operating in runaway regimes finds solid support in current theoretical modeling literature, while the fractal and deterministic extensions (Phase 3 and Phase 4) represent the least explored speculative cutting edge, charting the course for future publications in the domain.

7. Conclusions

The developed intelligent system has demonstrated the full engineering feasibility of applying Information Extraction, Text Mining, and unsupervised Machine Learning techniques for the epistemological analysis of complex scientific literature. The

implemented software architecture rigorously meets the criteria of modularity, maintainability, and responsiveness required by modern paradigms of Internet-based Intelligent Systems.

From a computing perspective, the integration of a multi-source adaptive crawling engine with an optimized regular expression-based processing interface enabled the seamless processing of a heterogeneous corpus of documents, normalizing different text formats and extracting explicit numeric entities structured in standard JSON schemas. The dual interface developed—an interactive CLI for system administration and a responsive Flask-based web dashboard with an asynchronous, threaded architecture—ensures excellent usability and operational flexibility in both local and remote control scenarios via REST APIs.

From a scientific and epistemological perspective, the Analytics Layer and the Learning Layer provided robust mathematical validation of the postulates of Approach Theory (AT). The calculation of Pearson correlations and the subsequent mapping of the Semantic Gap (consistently above 0.60) analytically confirmed the presence of a historical semantic gap between high-energy nuclear physics and traditional atmospheric lightning physics. At the same time, the K-Means clustering algorithm geometrically isolated the two communities as separate domains, demonstrating however that the transient models and relativistic RC circuit mechanisms predicted by TA lie precisely on the geometric transition boundary between the two clusters.

In conclusion, the system demonstrated that Approach Theory is not an isolated model, but rather the analytical link necessary to fill the semantic gap in the contemporary literature. The tool establishes itself as a robust automated knowledge platform, ready to be extended by integrating local Large Language Models (LLMs) for semantic refinement of the extracted relations.

8. Repository

https://github.com/DouglasRuffini/Sistemi_Intelligenti_Internet

9. Riferimenti

Ruffini, D. (2025), “Electron Approach Theory. A Damped Oscillation Model Based on Relativistic Effects and Space-Time Feedback”, (<https://www.gsjournal.net/Science-Journals/Research%20Papers/View/10173>).

(Ruffini, 2025)

Ruffini, D. (2025), "Electron Approach Theory. Fractal Extension. Mathematical formalization of absolute space and absolute time as a recursive projection, by means of a weak conjecture of a weak conjecture deriving from Goldbach's strong conjecture.", Press: The General Science Journal (ISSN 1916-5382) (<https://www.gsjournal.net/Science-Journals/Research%20Papers/View/10174>).
(Ruffini, 2025/A)